

SynapseFlow - Project Notes

Overview

SynapseFlow is a production-grade multi-agent LLM orchestration platform. It coordinates 8 specialized agents through a shared context blackboard, with 2-hop RAG retrieval, critique-synthesis pipeline, budget compression, and human-in-the-loop prompt optimization.

Architecture

FastAPI Gateway -> LangGraph Pipeline -> PostgreSQL / Redis / Gemini

Agents: Orchestrator, Decomposition, Retrieval, Reasoning, Critique, Synthesis, Compression, Meta.

The Orchestrator routes work using LLM decisions with deterministic fallback. Decomposition builds a typed subtask DAG. Retrieval performs 2-hop pgvector search. Critique scores claim spans; Synthesis applies RESOLVE/REMOVE/HEDGE actions.

Key Features

- Shared context blackboard with thread-safe state
- Never-silent-truncation budget policy
- Prompt injection screening at API boundary
- SSE streaming with PostgreSQL trace replay
- 15-case eval harness (BASELINE / AMBIGUOUS / ADVERSARIAL)
- Circuit breaker + semantic cache on LLM calls
- Prometheus metrics and structured logging

API Endpoints

POST /v1/query	Execute multi-agent pipeline
GET /v1/runs/{id}	Retrieve execution trace
GET /v1/runs/{id}/stream	SSE live agent events
GET /v1/runs/{id}/dag	Subtask DAG visualization
POST /v1/rewrites/approve	Human approval for rewrites
POST /v1/eval/run	Run evaluation harness
GET /health	Health check
GET /metrics	Prometheus metrics

Quick Start

```
git clone https://github.com/Prabhat-190/synapse-flow-.git
cd synapse-flow
python -m venv .venv && source .venv/bin/activate
pip install -e '[dev]'
synapse serve # http://localhost:8000/docs
synapse query 'Explain LangGraph orchestration'
synapse eval
```

Tech Stack

LangGraph, Gemini, FastAPI, Celery, Redis, PostgreSQL, pgvector, Prometheus, structlog, OpenTelemetry-ready instrumentation.

Evaluation

15 test cases across 3 tiers scored by an independent judge model on: correctness, citation quality, contradiction

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resolution, tool efficiency, and budget compliance.

Author

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